# The Sound of Teaching Music
# Experts' sound modulation for novices


**Atsuko Tominaga**[1]

Tominaga_Atsuko@phd.ceu.edu

**Günther Knoblich**[1] **Natalie Sebanz**[1]

[1] *Department of Cognitive Science, Central European University, Hungary*


## Introduction

- Experts tend to modulate their speech and action for novices (Brand, Baldwin, and Ashburn 2002; Fernald 1985).
- Expression is one of the crucial elements of expertise (Sloboda 2000).

**How do music experts modulate expressive performance for teaching purposes?**

## Methods

### Participants

**20 expert pianists** (years of practice: $M = 14$, $SD = 5.64$) were included for data analysis.

### Conditions

**Teaching condition:**

Pianists were asked to play the excerpt of music as if they were teaching it to students.

**Performing condition:**

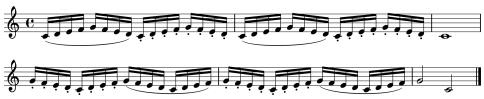
Pianists were asked to play the excerpt of music as if they were performing it to an audience.

## Predictions

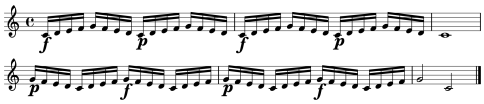
1. Pianists will play more slowly in the teaching condition.
2. Pianists will exaggerate legato and staccato in the teaching condition.
3. Pianists will exaggerate forte and piano in the teaching condition.

## Stimuli

**Articulation (legato / staccato) - Smoothness**



**Dynamics (forte / piano) - Loudness**



## Procedure

Step1: Practice with a designated expression

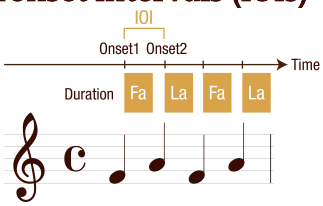
Step2: Perform the piece with the expression 8 times

- Repeat Step1 - Step2 for each skill in each condition (32 trials in total)
- The orders of the conditions and the skills were counterbalanced.

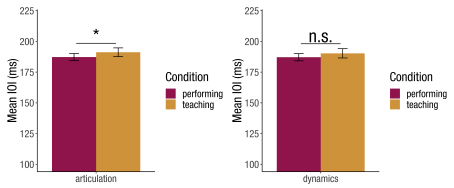
## Results

### 1. Tempo

**Interonset intervals (IOIs)**

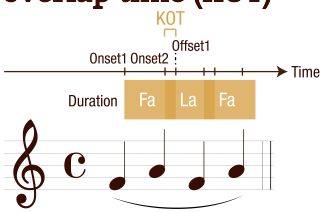



**Experts performed more slowly when teaching articulation, but not for dynamics.**

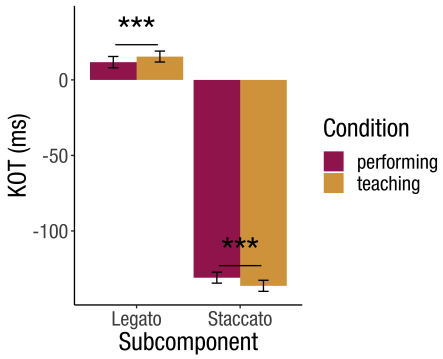



### 2. Articulation

**Key-overlap time (KOT)**



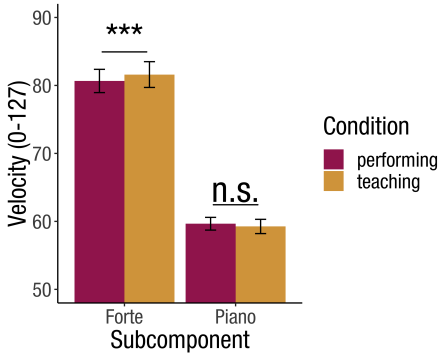

**Experts exaggerated legato and staccato while teaching.**



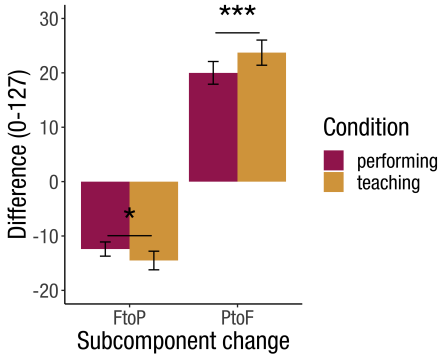

### 3. Dynamics

**Key velocity (KV; 0 - 127)**

**Experts exaggerated only forte during teaching.**




**They made a larger contrast between forte and piano for teaching purposes.**




Significance: * $p < .05$, ** $p < .01$, *** $p < .001$

## Discussion

- Can the findings be replicated with more realistic pieces?
- Future studies should examine which cues can be used to facilitate novice learning.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